

ActInf Textbook Group ~ Cohort 1 ~ Meeting 19 (Chapter 8, part 1)

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hello everyone

it's October 7th 2022

we are in meeting 19 for cohort one

and we're discussing chapter eight

so

there's a bunch of questions

and also a lot of things to talk about

let us just begin with looking at

chapter eight

so

does anybody want to uh

give any first thoughts

on active inference and continuous time

everything flows nothing stands still

Etc

yes

um

I not on that specifically okay but I

will have something to say about this

chapter I I worked on this topic

um during my my graduate time so what

should we get to about 8.3 I'll weigh in

excellent

um

any

comments like discrete continuous time I

know this is something we'll touch on

again and again

it's kind of like another low road High

Road

duality

with continuous Indiscreet time and I

think the textbook tries to give

equal time

and highlight the hybridization of these

models

in a way where realistically in the

literature most

are pure discrete models

and continuous time models

um have been almost like domain bound

certain areas like motor reflexes where

there's more of a tradition of

continuous models

are heavily

represented in them whereas decision

making models

tend to use the discrete formalisms and

that difference between the sort of like

continuous time traditionally motor

Associated and discrete categorical

traditionally decision oriented

models was addressed in the

um folk psychology

paper and live stream with Ryan Smith at

all

okay

um also I

just started kind of an overview got up

through eight three

of

just understanding and just trying to

lay out chapter eight so for any chapter

everyone's always welcome

[Music]

to just

make summaries and and add their own

notes

so we can look at the

questions

many of which are on the earlier part of

the chapter and so maybe we can look at

the

earlier part of the chapter more this

week and of course then have later time

uh

okay so 8.1

it's a single

paragraph

it's a complement to chapter seven which

was a discrete time approach towards

generative model construction

the focus here is on continuous State

space models which are well suited for

physical fluctuations sensor receptors

and continuous motion of effectors

they start with the case of movement

control 8.2

so

let's get familiar with the notation

the states

hidden states are going to be X

the data the observables are going to be

y

and then how States evolve over time

depends on a static variable V so that

could be a hidden or slower changing

cause or factor in the environment

then we have the Ω terms associated

with each of the data and the hidden

States change Through Time

and that variability

um is like the noise and this is like

the signal or the flow so it's kind of

the lengthen decomposition or language

approach towards

dynamical systems modeling

um

one interesting thing to note is first

that this is hugely foresaged in

um SPM

then

um also that the data

are a function of the states at that

time

whereas the equation for the hidden

States is a change in Hidden States

$\dot{x}$

note that action is absent from equation

8.1

this is because action is part of the

generative process not the generative

model

thought that was kind of an interesting

um

question

what does that mean that action is part

of the generative process and not the

generative model

surely action selection is part of the

generative

model

the cognitive process of policy

inference

active States

are part of the particular States

active states are blanket States

particular states are the generative

model

the generative process is referring

to the external States the hidden States

so is action part of both the generative

process and the generative model

because it kind of sits in between

intermediates

or is this referring

to the consequences

the generative model only deals with

those variables directly influenced by

States external to a Markov blanket
so is that saying that the generative
model is kind of like the complement
of the autonomous States action internal
States
or is this even more extreme
it's only dealing with the variables
directly influenced by States external
to a Markov blanket so then it must only
deal with sense States
which is like a narrow perceptual
control reading
and in that situ in that sense
action execution
is outside of the generative model
because the generative model is only
dealing with observations
does anyone have thoughts on how
generative model action and generative
process are being used Eric
I wonder if you can consider your own
actions part of what gets observed
under that framework
as far as I would uh the Apologetics for
the narrow View
would be exactly that
which is like you have
um uh proprioceptors in your elbow
so you don't observe action but you are
getting continuous updating of
perceptions that are extremely
influenced by action
but the hidden state of the world is
actually like the angle of your elbow
and so you're not directly observing
the hidden state of the world nor how it
changes all you have is the
proprioceptors and of course you engage

in policies to change
your proprioceptors bring them towards
alignment which is what we're going to
be discussing in the chapter
Ali I saw your hand race
uh yeah I wanted to mention exactly the
proprioception which you mentioned so I
lowered my hand
okay yeah great
but then um could you include your you
know the fact that you've got
signals going to your muscles your your
motor signals and motor neurons are in
there making your muscles contract can
that be part of your observation
to say well okay if this is happening
that these muscles are getting activated
then
sure enough the the arm moves an
appropriate perception changes so you
can say well from
from the outside of that box
that's all part of the generative
model
of what
of seeing you know but not part of
generating the those muscle commands
which is seeing what the effect of them
is
I wonder if that would be an acceptable
viewpoint
cool well we just we'll return to this
like where is action what is action
but I think it's just important to note
while we're early here
okay
so
continuing a little bit on

this part I thought was also a little
curious

impact so this is equation 8.1 we
mentioned equation A2 so first question
italics versus bold G and F

do you think that that matters

I don't think it matters

does anyone disagree

like do you think it's moving from like
a some sort from one space to another
space or do you think that's just the
typographical
error

I think it can be considered as an
errata

um one thing I did notice is that 8.3
and 8.4 the italics they call a
generative model an 8.4 they call a
generative process with without the
italics could that be the distinction
oh um okay eight eight point three and
four

yeah ah okay

it's a very good question well let's
just say italics an 8.1 and 3.

old in

8.2 and 8.4

um unclear if relevance difference

okay

foreign

question I mean if the meta model you
say frames it so if you consider the
niche to be a meta model then sure
the difference between equation one and
two other than the Bolding or not
is that V

the static variables have been replaced
with u actions

so this takes us from like the sort of
SPM
where you'd be like you have
neuroimaging data y
as a function of underlying neural
States X and then neural States changing
through time are a function of neural
States at that time and some conditional
Factor
that you're testing for like whether
they were exposed to treatment water
treatment two V
so this is still a um
partially observable dynamical model
but then it gets moved very neatly into
the control theoretic and it
policy selection is inference Space by
just saying yeah those slowly changing
factors that influence how the hidden
States change
we control those
or we have some ability to influence
there
okay
yeah so this could be a cue towards the
Bold difference which is like the
functions G and F
are not necessarily the same as those
used to define the generative process if
they said not necessarily the same as
those used to define the general process
bold G and F then it would be
unambiguous
but it's a little ambiguous
we do not need to explicitly write down
the Dynamics of action in the generative
model
okay

they emerge from a choices made for
terms in equation 8.1

we start with a simple sort of
generative model

so G which is the function that's
generating the data is going to be
 X

f
the function of how X changes Through
Time

is a function of X and V the slower
change in causes

and it's the difference between those
two

the hidden State represents the expected
value for the data

or in other way say that would be
equation three says that the expected
value for the data are the hidden state
but does the same

and it's a point attractor

or in the Bayesian mechanics we could
say it's mode

matching

by a tractor

um

uh yeah

yes the attractor here is the mode

tracking

that is being referred to here

when there's a difference between the
hidden State and the hidden cos V

then

it moves towards it so it's just moving
towards a single fixed Point attractor

now the generative process is defined
so it looks almost identical

except these are replaced with u 's

and this differential tracker
this kind of PID control Motif is
replaced with
action
it's an expression of the equilibrium
point hypothesis
which treats motor control as enacted by
reflex arcs that draw limbs towards
equilibrium points set by descending
motor signals
in uh and then also this is related to
the skilled performance
live stream 23.

the scheme does not require
specification of inverse models
that are widely used in other
formulations of motor control

8.5

makes a slightly more sophisticated
model

that's based around not just position
because of course you can't just like
pick up your finger and move it to a
different position

really the action affordances are in the
space of changes in velocity which is to
say acceleration

so this is kind of a classical
mechanical perspective on motor control
and it also points towards uh
generalized coordinates of motion

and that kind of PID control
and the Bayesian mechanics and so on
okay

now we get to 8.3

continuous time formulations are well
suited to characterization of movements
well certain kinds of movements on the

chess board maybe not
but some kinds of movements sure Eric
yeah I like to point out with um
equation
8.5
um just above that
where he's got the they've got this
um spring Mass
so
um
that um that system oscillates forever
right it's got a spring and a mass
and there's no dissipation of energy
so it's it's a step up I guess better
than
um in a more advanced than
um equation 8.3 in which you have
instantaneous acceleration so they
recognize okay you can't have
instantaneous acceleration but there's
no there's no uh damper here so normally
the the the true equilibrium point
control formulation is viewed as a
spring Mass damper system
and
um
that way if you if you remember from
physics class you know depending upon
your set your your mass the spring
tension and the amount of damping you
can get over damping or under damping
and then you can get kind of an optimal
settling to the uh or uh approach of the
set point with maybe just a little bit
of overshoot or no overshoot but if you
don't have a damper in there then you it
just never it never cut cells in your at
the equilibrium point

but um

so just to carry this a bit further um

I I have to would have to to disagree

with this statement that continuous time

formulations are well suited to

characterization of movements because

that's that's assuming that a the

equilibrium point control model is a

good characterization of movement

I worked on this in grad school for

about two years when I first got to grad

school we worked on

um

the equilibrium point control

model in I work in the BT Lab at MIT

and we we had an apparatus

um that mimicked the apparatus that was

used in monkeys we had one that we used

on ourselves as human subjects which

would um allow you to move your arm

and then it would kind of give you a

little jolt a little a little disruption

to measure the amount of stiffness and

in your movement as you're moving your

arm

as a way of trying to deconstruct or or

work back the parameters of the

equilibrium point control

uh model for for movement and it's

attractive because

um as they say in this in the chapter

um well you don't have to have any very

sophisticated

um knowledge of motor planning

or inverse model of the of your your

plant the system

in order to all you do is you say well

this is where I want the arm to go and

passively uh it takes care of itself
perhaps through
um a feedback loop in the you know
through the spinal cord which I guess is
what 8.1 has
um it's interesting that that there's a
paper from 2009 uh 11 and Feldman that
is you know reviewing this but um our
conclusion was pretty much that
equilibrium point control
for motor motor control is just it
really is is not a good model it doesn't
work
um
in order to make it work you had to have
all kinds of play all kinds of
shenanigans like well you're not just
have a fixed set point you might
overshoot and then you bring your set
point back and you actually have to
control the stiffness during movement in
order to account for you know observed
data but still it falls short because it
because many if not most human movements
or any you know motor movements um are
ballistic
so it's not that you
set some conditions and then your your
spinal cord and motor and muscles act to
fulfill those conditions that you sat
once from the top from your brain
what really happens is
you're you're descending signals
anticipate
a whole trajectory or a whole plan in
advance of how we're going to accelerate
and then we're going to have to
decelerate

um

and so it's it's the motor planning kind
of knows what what your plant is going
to do it doesn't wait for the plant to
do something and then have to respond to
that in kind of a passive way at the
lower levels

um so it's just like you know if you
play the piano you're not waiting for
your finger to say am I far enough down
on the key

you you kind of you learn how hard you
have to press in and when you when you
start to the release before anything
happens I mean it's our movements are
very very fast

and and um so there's really no getting
around having to model the plant and
doing doing this sort of inverse
modeling that they're they're concerned
about

um in in order to account for how how
any creature really moves or anything
you know above an insect and probably an
insect who's

um undoubtedly of it's a motor plan so
um so I really find that this this
chapter

um is

is is really not

um I don't think it advances the cause
of

um active inference it doesn't really to
me um

it doesn't really fit with with with
with with how about a control really has
to work and doesn't add anything to to
the theory of it I mean it's using an

old theory that doesn't work
thanks for all those points
that's my statement
that's it that's the tweet
um well one uh live stream that other
than 49 which has just been like
amazing from the physics angle we're
preparing also for 50. which is using
it's also by
um
felden Barrett and others
and it's specifically on interceptive
reflexes so less exploring like the
spinal reflex arc and more on the
vasopressive interception
but using
those kinds of still classical
very
um
differential equation based
equilibria control
and showing that those types of
formalisms can
recapitulate like allostatic Behavior
it's always difficult to know like is
that a very narrow and fragile range of
parameter space
is this a descriptive model that's
making unique predictions
let's let's look at figure eight one
because it's speaking to um
the
generative model generative process
distinction that was mentioned earlier
and it was addressed in a question
how is this representation of the reflex
arc similar or different to what was
being used in figure five

or in chapter five so in chapter five
here was the summary statistic
or summary graphical abstract there was
a lot more focus on this kind of like
um policy selection and so on but we we
um did have
this kind of a reflex arc
why
e
and so on okay
so now
we're in 8.1
f
is the action selection
component
so here is that simple attractor being
shown
where
it's just X is going to be converging to
 V
and this is the change in
external States
so here we have μ of x
in an equation
in the lower one
is this one referring to this or is that
kind of bleeding out from the bottom
panel
because here's U of X and here's you
have μ of x
here it looks like μ of X is a
prediction modulator
on a differential
so it's like if the observation is where
you want to be then there shouldn't be
any action
but as the observation diverges from
where you want to be

you have a Precision modulated action
is that active inference
or is that just
a notation
um a different notational representation
of

classical equilibrium point motor
control theory

do those models integrate
sensory and action
together

I mean in one sense they definitely do
nobody was ever generating reflex arc
models without some
sensory feedback variable
so that's open important question how is
this is this didactic with respect to
active inference now just kind of saying
yeah one special case of active
inference is

um the traditionally understood and
limited equilibrium point model of motor
control

or is it like this is our new way of
doing Motor Point control

okay just kind of
continuing on

all right Precision attention and
sensory attenuation

so here's another

um

curious note so earlier we we uh
discussed whether motor control was or
wasn't a natural setting for continuous
models

in the Box it won they write we
addressed this the importance of
precision in chapter seven but it's

worth recapping its role in continuous
time systems in many ways this concept
of precision
is more naturally addressed in this
setting

as the π variable appears a direct
consequence of Laplace approximation
okay

but can't you do a LaPlace approximation
in discrete systems as well

LaPlace approximation is just mode
determination followed by the
parameterization of a parabola
to create like a gaussian or at least a
concave

uh
optimizable

and
two parameter approximation to an
arbitrary distribution

I don't exactly see
or is it referring to the LaPlace
approximation is itself a continuous
function

but you could apply a LaPlace
approximation to a categorical
distribution as well

so in what ways is precision more
naturally addressed
in a continuous model

is it that Precision is a continuous
variable

like Precision isn't an integer valued
variable

so

we can just leave that one

okay but so we went into eight three
dynamical systems kind of looked at the

spinal cord again

but it may or may not just be a sort of

like trivialized or didactic

representation of extremely traditional

representations of motor control

then box one Precision attenuation

attention sensory attenuation all right

now we get into the lot kevaltera

Dynamics

so

what are some formal and qualitative

implications of the use of Latka

Volterra Dynamics and Lorenz systems

so I think there's a few interesting

angles that that us here now but others

just more

um broadly may be interested in

so one is lockabletera ties in of course

with ecology

and population ecologies

and so that is also very interesting to

consider in terms of like winnerless

competitions and neurodynamics and that

is basically the analogy which is like

just like even if you have a Negative

Edge between the predator and the prey

for any persistent which is to say

oscillatory or manifold attracting

population

at some point like you have to let the

prey regrow

so the analogy in Neuroscience that's

fleshed out in SPM is like if you have a

brain region even with a unilaterally

negative influence on another region

at some point it has to like let off the

break cyclically

so lockabletera summarizes these kinds

of like winnerless dynamics that can
give oscillation and time structure
including with like complex Dynamics and
recurrent Dynamics limit attractors all
these like kinds of dynamical processes
can arise from a pretty
simple
starting point
and uh here we see
manifolds and like limits on the
projection because it's a
three-dimensional space plant herbivore
carnivore or that could be the um blood
oxygenation
of three brain regions in SPM and this
is their neural trace or their inferred
activities and so then you can find like
manifolds of lower dimensional activity
and then also one um kind of related
note
is the Volterra series
and this is a pretty nice just technical
note which is Taylor series
does expansion
of
arbitrary functions from a starting
point
and the Volterra series is able to
include like a window of inputs also
known as the memory effect
and so it's discussed in
um SPM it has a continuous time and a
discrete time variant
and it is just a kind of Kernel
estimator
that can be used for dynamical processes
input output dynamical processes with
memory

so it's kind of a classic in that area
and I think even in some live stream
maybe 45 I asked Tristan about the
lockable terror models and about like
why
like where did we move away from them or
something like that but that'd be a good
answer to kind of come back to
all right so
um now those kinds of lockable Terror
Dynamics are going to be applied
um
monochromatically as ever
to active inference
and that was one question which is uh it
didn't finish writing how is the
lockableterra model being used in the
case of handwriting
foreign
and herrera's paper
which I just brought in but I didn't
really investigate deeply
um
the left
there's two different applications here
left
eye blink conditioning
so here it looks like a conditioned
stimulus unconditioned stimulus
where there's like
where something happens and you blink
as we've mentioned previously like it's
a little bit difficult to tell exactly
what these expectations are of or
anything like that
but
this is like
some kind of stimulus response

blink blink once if no Twice of yes once

if I land twice if I see

and then on the right side

sequential Peaks using an attracting
point

but selecting the specific attractor on
the basis of which population a
lockabletera system is currently highest

this leads to sequential visiting of
each points giving rise to a type of
handwriting so this is kind of

interesting and I I know for Jacob and
Ali also with our mode tracking path
matching discussion Bayesian physics

um so it's like

um I um I must

it's Bart Simpson at the chalkboard

this is I must rewrite the sentence I

must rewrite the sentence I must so your
expectations are that you're going to be
writing this sentence

and then

depending on which of those expectations
are on top

that is going to set

the motor

imperative

so it'd be like if you were writing the
letter

um D and it'd be like well it's like

first there's like an arc down and to
the right and then there's an arc down
to the left and then it's a straight

line up and then it's an arc down to the
right and then it's an arc down to the
left and then it's a straight line up

and so here's kind of like doing this
writing

as it's
picking a mode pursuing the mode
and I think it's fair to say uh jakov
early please
add any more if you feel like it but
like
this is like
shaking off the slumber
of the classical physics and still
having one foot in both worlds
and the three faces of Bayesian
mechanics
are a much more advanced way to discuss
this issue
yeah Jacob what do you think
I was just going to say that
um
this
well not the case with writing
specifically but uh
the case of motor uh motor control
was discussed in Thomas parr's lecture
um at the CPC 2022 which I don't know if
it was recorded uh actually I think it
was
uh but I don't know if it's public yet
um but it's interesting that this
example is in the continuous Time
chapter because you could think of it as
a nested generative model with both
discrete and continuous uh
uh factors like if you partition it into
three decisions like uh
for a straight line up then an arc down
then a second Arc down
um those are three discrete decisions
that are at the top of the
of the hierarchical generative model

with the
finer grain motor control below
thanks also just here's that classic
three faces figure
mode tracking this can be seen as using
a log of ultra pseudo random number
generator
to spit out higher level predictions
about which mode should be pursued
then as to the Dynamics of how that mode
is approached
one can either use the equilibrium
points
methods discussed in this chapter or
the more advanced Bayesian physics
methods to describe that in terms of
gauge Theory and so on is that fair to
say
oh
is this it jakub thank you Ali for
sending it
uh it's it's not the same it's not the
same one but it looks like a lot of the
slides are
the the same I think the thing that I
was referring to
I'm just trying to find the slide
oh yeah I think it might be
minute 41 okay
I I really can't wait till there's some
open source webcam software
to do
isicade analysis and um
pupil diameter
I don't think it would be that
intractable maybe you'd have to be kind
of close or you'd have to use a 4k
camera or something

but is there any software like that
available I've never come across it
just like
move the mouse to where I'm looking and
then click when I blink twice
isn't that kind of like that exists that
does exist but it but what is it called
or where is it being used

yeah

I know it's being used uh for
um people with physical impairments who
can't control the computer

uh

yeah otherwise yes I don't I haven't I I
haven't seen I haven't seen uh like an
analysis of people dilation

but eye tracking is definitely a thing

okay maybe we can explore it

um all right so yeah the blinking
example was a puff of air so it was just
like a beep and then a puff of air
and so then they learn that the they
learn to preempt the air Puff

okay

learning in continuous models eight two
continuous time domain accumulating
evidence this works if we treat data in
a series of small time intervals

oh no don't discretize

um this relates I think to some of the
finer points on like which stochastic
calculus is used like the Edo calculus
in the Bayesian mechanics

um

we won't go into it unless anybody
really wants to say more but this looks
like important just to learn about the
integrals and the gradients

okay pomdp has largely superseded the
use of the generalized lockable
terrorist systems
inactive inference applications
and that was kind of I think related to
asking first about why things moved away
from the lockable terror formulations
Lorenz system
also explored in stochastic chaos Markov
blankets live stream 32.
it's it's almost it's dealt with as
minimally as you could possibly deal
with the Lorenz system
in eight seven okay
eight four generalized synchrony
so multi-aging inference starts coming
into play as well as agent environment
synchronization and we actually like
talked a little bit about the bird song
in the previous hour
um now the Run system comes back
and yes it's dealt with like sort of
in the 2015 frith and friston
paper active inference communication and
hermeneutics
and that's where the synchronization
manifold is kind of displayed and then
more recently in
um
live stream 32
is um and this one has a DOT three I
think this might be the only dot three
as of this point
I think Connor Hines could only come at
a later date yeah
so these are good discussions
on the lorentz tractor a couple Duran
systems

this is kind of um very similar if not identical to the uh bird song synchronization manifold in first and forth 2015.

birds birds birds

um

bird synchronization isn't real

then hybrid discrete models

um

hybrid mix models

not that kind of mixed model

allow inferences about sequential action

plans and translations of those

decisions into movements through a

continuous model

so

in that respect

it's extremely linked to livestream 46

active inference models do not

contradict folk psychology which is all

about that continuous time motor active

inference

discrete state-space decision active

inference Mai and Dai and using those

motor and decision active inference

models

while lower level model

in the dashed box

is the same form as the other models

um in in other words it's continuous

because we have V instead of P_i

for the causal factors and F instead of

B

and G instead of a

and uh

ETA instead of

G

is it

some mapping between well in this case I
think just looking at it a little more
narrowly how
um outcomes
map
to slower causes of the world
so here this is a discrete model up top
with a G oh yeah that was one era that
we found
free-floating G
can't live with it can't live without it
but it should be connected to the pi of
course
um
there's a discrete time model on top
and then it is kind of cascading its
outcomes
down
to these continuous time models
so one could think about this like the
lock of Volterra handwriting example
kind of could it's not the exact
structure of this but it's related
okay
um this is
um something I know Jacob has thought
about
um how discrete and continuous models
um interface
because we've talked about how like the
temperature could be continuous and the
thermometer could be continuous but then
the internal generative model could be
like is it hot or not
so another question of how to how to
um
move from generative models
within between different parts of the

model where where or what kinds of
transitions between continuous and
discrete State space is possible
okay

here's a continuous

um isocate

uh ocular motor paradigm

where the continuous lines

are still confusing but at least they
are Justified

eight three

combining categorical and continuous
generative models

outside of active inference has

primarily been framed to the lens of
clustering

the aim is to assign data points into a
discrete cluster

foreign

sure about that I think clustering is
one discretization clustering and
thresholding is one discretization
approach that's been used

also just binning

seems like a simpler method

how do you say a continuous distribution
of one through ten and make it
categorical

just bin it one through ten or you could
cluster and do thresholds so then you
could have

um cluster one with 9.7 to 9.9 and then
9.7 to 9.5

um

and then eight six

there's a big table

here's the songbirds

this author isamura I believe is the one

who will join

per 51.

in November and I think that'll be like
our last probably one of our lost paper
streams for the year

we have some we have just this last
discussion with Dalton next week
and then interception and then the
canonical neural networks perform active
inference papers

but isomers worked with power and first
and others oculomotor reflex eye Pursuit
psychosis illusion cicade action
observation attention hybrid and
self-organization models

well

with four minutes left

kind of an interesting chapter

but I can only know it how I'm

seeing it and I

have serious questions about what it
does in the book

Eric

um one thing I was puzzled by and maybe
you guys can help me out with on the
bird the bird song examples were they
implying that um

Bert starts singing

another bird picks it up synchronizes
using these Dynamics
and

um the proof of that is this other bird
will just step in and take over and the
first bird can stop

I don't believe I've ever ever heard
that happening are they suggesting that
that happens or or what

um

okay I agree like whether it's a improv
whether this is a bird improv
conversation in this case they're
singing from the same hem sheets
so it's like somebody should be singing
the bird anthem
and now we're going to be engaging in
this sort of Quasi dialogue
where we're going to be
reducing our surprise about hearing the
song that we expect to hear
which goes like you know
insert your favorite song there
um
and then the phenomena that gets
highlighted
is the way in which
um one bird is singing
and then as it's singing its Precision
starts to drift
and then
it stops singing and then the other bird
finds itself picking up at the same
point
to reduce its surprise
about that song being sung at the time
but
I think especially with it with a bold
title like land claim to hermeneutics
more broadly
singing from the same
pre-coordinated hem sheet
is not the same thing as information
updating or even just pure expressivity
so I I think it
it leaves
gaping areas for people to actually
develop

because it's like well we have what
don't we have databases of like millions
of bird songs
and natural soundscapes
so like we need some bird watching
active inference practitioners
who want to actually develop
the bird song Motif because otherwise it
is too easy and ironically becomes
singing from the same hymn sheet oh well
hermeneutics has been addressed in 2015
synchronization manifold 2015.

um

whereas this is like a very Fringe case
of communication
that is based upon

um

most charitably High reliability
execution of
pre-ordained performance
which is a tiny Edge case of
communication

you know my understanding of bird songs
is you know the the

um baby birds here here the with their
parents saying you know they've got some
predisposition to to sing a certain kind
of song and um for most birds you know
there are some that are very adaptable
you know parents parrots you know stuff
like that but most birds

um learn the accent and the or the um
the dialect that they are raised in and
that's the song that they always sing
um and there's some variation on it I
mean maybe they'll stop Midway through
but what what are the true facts about
I what I saw to be a claim here where

birds will pick up
from each other or one leaves off does
that really happen I I I I'm not
familiar with that happening I've never
I don't think I've ever heard it
happening and what my impression the way
most birds sing is they'll sing their
song all the way through
and there might be a column response
another one might
might pick up uh you know might answer
but I don't think I've ever here
synchronizing of songs
um but I might be just not well enough
informed about the Ecology of those
things
yeah very interesting um
questions
and the whole area with game theory of
song
like
it's beneficial that somebody warns
about the Predator but then you've
revealed your location when you sing and
things like that like that's the kind of
decision making
and um uncertainty resolution and risk
evaluation that I'd imagine
in ancient-based bird song model
wood oriented towards but those are some
cool ways for us to go so
um next week
I I guess this is kind of a shorter
chapter and it has some Arcane symbolism
but also seemingly very skippable
symbolism
so I'm glad that we could get through
the whole thing

and um will return to any further
thoughts on eights
and then maybe we can have more of a uh
seven plus eight synthesis
as or anyone can look into any of these
citations key advances
as we head into
um like chapter nine
on empirical data
which is not super long
followed by 10
which is just a restatement
so
um onwards we go thanks y'all bye
thanks everyone